LEARNER GUIDE

Partitioning in more than one way

Place Value · Lesson 5. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

To **partition** a number is to split it into parts that add back to it. 3.47 is 3 + 0.4 + 0.07, but one one is worth ten tenths, so you can **exchange** 1 one for 10 tenths: 3.47 = 2 + 1.4 + 0.07. Ten of any column are worth one of the column to its left, so breaking and grouping change the split but never the number.

What you are learning to notice

- Breaking: 1 counter becomes 10 counters in the column to its right.
- Grouping: 10 counters become 1 counter in the column to its left.
- After any exchange, the parts still add to the same number.
- 3.47 can be 3 + 0.4 + 0.07, 2 + 1.4 + 0.07, 34 tenths + 7 hundredths, or 347 hundredths.
- When a column is not fully used, what is left over joins the next column: 5.8 = 4 + 1.8.

Moving through it, one step at a time

To find a missing part, put the number in a chart. Take out the parts you are given, breaking a counter into ten whenever a column runs short. What is left is the missing part. To check a split, add the parts column by column and compare with the number.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Breaking a one

Show that 3.47 = 2 + 1.4 + 0.07.

3.47: counters in each column

Ones		Tenths	Hundredths
3	•	4	7

Break 1 one into 10 tenths

Ones		Tenths	Hundredths
2	•	14	7

The numbers count counters. 2 ones, 14 tenths and 7 hundredths: still 3.47.

1. 3.47 is	3 ones, 4 tenths, 7 hundredths
2. Break 1 one into 10 tenths.	2 ones, 14 tenths, 7 hundredths
3. 14 tenths is	1.4
4. Read the split.	2 + 1.4 + 0.07

3.47 = 2 + 1.4 + 0.07. Taking 1 one away and adding 10 tenths changes nothing.

Example 2 — A missing part

52.6 = 4 tens + ? ones + 6 tenths

52.6

Tens	Ones		Tenths
5	2	•	6

Break 1 ten into 10 ones

Tens	Ones		Tenths
4	12	•	6

The 10 new ones join the 2 already there: 12 ones.

1. Only 4 tens are used, so break the fifth ten.	10 ones
2. Add the ones already there.	10 + 2 = 12
3. Check: 40 + 12 + 0.6	52.6

12 ones.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (tenths and hundredths (Lesson 4)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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